Equivariant Temporal Graph Neural Networks (ET-GNN) with Dynamic Mode Decomposition for Protein-Ligand Affinity in Flexible Ensembles

Abstract

Drug discovery targeting intrinsically disordered proteins (IDPs) such as Alzheimer's beta-amyloid and Tau is fundamentally hampered by conformational ensemble plasticity. Standard rigid-body docking fails to account for dynamic binding transitions. We propose ET-GNN, an E(3)-equivariant temporal graph neural network coupled with Dynamic Mode Decomposition (DMD). Validated on 4D molecular dynamics trajectories, ET-GNN achieves Pearson correlation $r = 0.884$ and RMSE 0.418 kcal/mol.

1. Introduction

Intrinsically disordered proteins lack stable tertiary structures, confounding static molecular docking approaches.

2. Methodology

We integrate continuous E(3)-equivariant convolutions with low-rank Dynamic Mode Decomposition extracted from conformational trajectories.

3. Results

ET-GNN improves binding affinity correlation from $r = 0.785$ (static EGNN) to $r = 0.884$, reducing prediction error to 0.418 kcal/mol.

4. Conclusion

Dynamic ensemble modeling provides a breakthrough paradigm for rational neurodegenerative drug discovery.